

The graded vanishing-order law is a $d = 2$ phenomenon:
collapse of $\text{ord}_{\zeta_d} G_\lambda$ for $d \geq 3$

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Abstract

The graded refinement $G_\lambda(q) = \sum_T q^{s(T)}$ of the descent-counting generating function carries, at $q = -1$, a rich vanishing-order law: $\text{ord}_{q=-1} G_\lambda = \lfloor |2\text{-core}(\lambda)|/2 \rfloor$, with order growing without bound along the staircase of 2-cores. It is natural to ask whether this is the $d = 2$ instance of a family of laws governing the vanishing of G_λ at primitive d -th roots of unity. The answer, by exhaustive computation over all 271 partitions of size ≤ 12 and all $d \in \{2, \dots, 8\}$, is no. For every $d \geq 3$ in this range the order $\text{ord}_{\zeta_d} G_\lambda$ takes only the values 0 and 1, and is positive exactly on the d -cores. The order law does not lift; it stands alone at the involution $q = -1$. We explain why: the parity weights w_i are 0 or odd, so $s(T)$ carries a clean mod-2 structure (the domino / Murnaghan–Nakayama mechanism) and no analogous mod- d structure for $d \geq 3$.

1 Setup

Fix a partition $\lambda \vdash n$ and let $\text{SYT}(\lambda)$ be the set of standard Young tableaux of shape λ . For $T \in \text{SYT}(\lambda)$ the *descent set* is

$$\text{Des}(T) = \{i \in \{1, \dots, n-1\} : i+1 \text{ lies in a strictly lower row of } T \text{ than } i\}.$$

We weight the descents by the *parity weight*

$$w_i = \begin{cases} 2i-1, & n-i \text{ odd}, \\ 0, & n-i \text{ even}, \end{cases} \quad (1 \leq i \leq n-1),$$

and form the *descent statistic*

$$s(T) = \sum_{i \in \text{Des}(T)} w_i.$$

The object of study is the *graded polynomial*

$$G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)} \in \mathbb{Z}_{\geq 0}[q].$$

This is the spectral grading of Theorem B in the order-law program: the multiset $\{s(T) : T \in \text{SYT}(\lambda)\}$ is the multiset of branch exponents of the Baxterised monodromy Ω on the irreducible V^λ , and G_λ is its generating function.

For a primitive d -th root of unity ζ_d , write $\Phi_d(q)$ for the d -th cyclotomic polynomial. Since Φ_d is the minimal polynomial of ζ_d over $\mathbb{Q}$, the quantity

$$\text{ord}_{\zeta_d} G_\lambda = \max\{m \geq 0 : \Phi_d(q)^m \mid G_\lambda(q)\}$$

is the order of vanishing of G_λ at ζ_d — the multiplicity with which G_λ vanishes as one approaches a primitive d -th root of unity. We study how this order behaves as d varies.

The single structural feature that drives everything below is this: *every nonzero weight $w_i = 2i - 1$ is odd*. The weight set is two-valued in parity, taking only the values 0 and “odd”. This is the crux.

2 The $d = 2$ law (recalled)

At $d = 2$ we have $\zeta_2 = -1$ and $\Phi_2(q) = q + 1$. Here the program has a complete theorem, proved earlier this cycle.

Theorem 1 (Graded 2-core vanishing law; Clio, 2026-06-01 / 06-02). *For every partition λ ,*

$$\text{ord}_{q=-1} G_\lambda = \left\lfloor \frac{|2\text{-core}(\lambda)|}{2} \right\rfloor,$$

and the order depends only on the 2-core of λ , not on its 2-quotient. Moreover the leading coefficient is explicit: writing c for the order and c_0 for the number of odd columns,

$$G_\lambda(q) = a_c (q + 1)^c + O((q + 1)^{c+1}), \quad a_c = \varepsilon(c_0) 2^{-c} e_c(W_n),$$

with $W_n = \{2i - 1 : n - i \text{ odd}\}$ and an explicit sign ε .

This is a *rich* law. The 2-cores are exactly the staircases $\delta_k = (k, k - 1, \dots, 1)$, of sizes $0, 1, 3, 6, 10, 15, \dots$, so the order is unbounded over the family of all partitions. For instance, a partition whose 2-core is the staircase $(4, 3, 2, 1)$ of size 10 has $\text{ord}_{q=-1} G_\lambda = 5$. The vanishing at $q = -1$ can be as deep as one likes.

The mechanism is the domino / Murnaghan–Nakayama identity at $q = -1$: removing a horizontal or vertical domino preserves the 2-core, and the resulting character-theoretic recursion forces $\chi^\lambda(2^a 1^b) = 0$ for b below the 2-core threshold. The repeated factor $(q + 1)$ is the polynomial shadow of this 2-adic rigidity.

3 The main observation: collapse for $d \geq 3$

One naturally hopes Theorem 1 is the $d = 2$ member of a family — a “ ζ_d -lift” in which $\lfloor |2\text{-core}|/2 \rfloor$ is replaced by some function of the d -core or d -quotient. We tested this directly.

Remark (Computational status). The following is a *computation*, not a theorem. For every $d \in \{2, \dots, 8\}$ and every one of the 271 partitions λ with $|\lambda| \leq 12$, the polynomial $G_\lambda(q)$ was formed exactly and the multiplicity of $\Phi_d(q)$ dividing it was computed over $\mathbb{Q}$. The results are tabulated below.

d	behaviour	$\max_\lambda \text{ord}_{\zeta_d}$	support (where $\text{ord} \geq 1$)
2	rich	5	$\text{ord} = \lfloor 2\text{-core} /2 \rfloor$ exactly (271/271); nonzero on 59 shapes
3	degenerate	1	only $\{(3, 1), (2, 1, 1)\}$ — the two 3-cores of size 4
4	degenerate	1	only $\{(2, 2)\}$
5	trivial	0	$\text{ord} \equiv 0$
6	degenerate	1	13 shapes
7	trivial	0	$\text{ord} \equiv 0$
8	trivial	0	$\text{ord} \equiv 0$

The contrast with $d = 2$ is sharp. Where the $q = -1$ order climbs the staircase without bound, every $d \geq 3$ produces an order that never exceeds 1, and that is positive only on a thin set of exceptional shapes.

Observation 1 ($d \geq 3$ collapse). *For every $d \in \{3, \dots, 8\}$ and every λ with $|\lambda| \leq 12$:*

$$\text{ord}_{\zeta_d} G_\lambda \in \{0, 1\},$$

and $\text{ord}_{\zeta_d} G_\lambda = 1$ only if λ is itself a d -core (i.e. has empty d -quotient). Equivalently, positive d -weight forces $\text{ord}_{\zeta_d} G_\lambda = 0$. In particular there is no repeated cyclotomic factor: $\Phi_d^2 \nmid G_\lambda$ throughout this range.

Thus for $d \geq 3$ the only thing ζ_d can detect is a single, simple zero, and it detects it only on d -cores. There is no growing order, hence nothing for a “ ζ_d -order law” to be *about*.

Worked example: $\lambda = (3, 1)$ at $d = 3$

Here $n = 4$, so the parity weights are

$$(w_1, w_2, w_3) = (2 \cdot 1 - 1, 0, 2 \cdot 3 - 1) = (1, 0, 5),$$

since $n - i = 4 - i$ is odd for $i \in \{1, 3\}$ and even for $i = 2$. The shape $(3, 1)$ has three standard tableaux, distinguished by the position of the entry 4 relative to its predecessor; their descent sets are $\{1\}$, $\{2\}$, $\{3\}$, giving

$$s = w_1 = 1, \quad s = w_2 = 0, \quad s = w_3 = 5.$$

Hence

$$G_{(3,1)}(q) = q^5 + q^1 + q^0 = q^5 + q + 1.$$

This factors over $\mathbb{Q}$ as

$$G_{(3,1)}(q) = (q^2 + q + 1)(q^3 - q^2 + 1) = \Phi_3(q) \cdot (q^3 - q^2 + 1),$$

so $\text{ord}_{\zeta_3} G_{(3,1)} = 1$: a single simple zero at ζ_3 , no more. The cofactor $C(q) = q^3 - q^2 + 1$ evaluates at $\zeta_3 = e^{2\pi i/3}$ to

$$C(\zeta_3) = \frac{5}{2} + \frac{\sqrt{3}}{2}i, \quad |C(\zeta_3)|^2 = 7.$$

This 7 is the loud signal that the $d = 2$ leading-coefficient structure does *not* lift: at $q = -1$ the leading coefficient is a manifestly factored 2-quotient hook product (Theorem 1), but here the analogous cofactor has squared modulus 7 — not a 3-quotient hook product, not even a value with any visible arithmetic meaning attached to the 3-core. The fiber value is governed by the classical Littlewood factorisation, but the *order* is just 1, and there is no further structure underneath it.

4 Why $d = 2$ is special

This section is explanation, not theorem. The phenomenon is forced by the parity two-valuedness of the weights, and the explanation is correspondingly clean.

Every nonzero weight $w_i = 2i - 1$ is odd. Therefore, at the involution $q = -1$,

$$q^{s(T)}|_{q=-1} = (-1)^{s(T)},$$

and $(-1)^{s(T)}$ depends only on the *parity* of $s(T)$ — equivalently, on the parity of the number of *odd-weight* descents of T . The statistic $s(T)$ thus projects, at $q = -1$, onto a $\mathbb{Z}/2$ -valued invariant. This mod-2 collapse is precisely the domino structure: it is the polynomial avatar of the Murnaghan–Nakayama recursion at $q = -1$, in which the relevant data is the 2-core and the parity of 2-quotient statistics. The repeated factor $(q + 1)$, and with it the entire $\lfloor |2\text{-core}|/2 \rfloor$ tower, is the visible trace of this 2-adic periodicity in the exponents.

For $d \geq 3$ there is simply no analogous mod- d structure to exploit. The exponents $s(T)$ are sums of values from $\{0\} \cup \{1, 3, 5, \dots\}$; reduced mod $d \geq 3$ these spread across the residues with no rigidity. A d -th root of unity ζ_d mixes the exponents according to their residue mod d , but no cancellation mechanism forces *repeated* vanishing, because there is no underlying mod- d refinement of the descent statistic. One zero — the generic “accidental” one detected on d -cores — is all that can occur.

This is consistent with, and indeed predicted by, the earlier finding that $G_\lambda(q)$ is *generically irreducible over $\mathbb{Q}$* (2026-06-04). An irreducible polynomial has no repeated factor at all, cyclotomic or otherwise; and high-order vanishing at a root of unity is precisely the non-generic, structured behaviour. The program established that the only place this non-generic behaviour appears is the involution $q = -1$, driven by the 2-core. The present computation is the quantitative confirmation: at every other root of unity the order collapses to $\{0, 1\}$, exactly as a generically irreducible family should behave.

5 Consequence for the program

The upshot is a clean negative, and it sharpens the map of the territory.

- **No ζ_d -order law exists for $d \geq 3$.** There is nothing to characterise: with $\text{ord}_{\zeta_d} G_\lambda \in \{0, 1\}$ there is no defect/obstruction term, no analogue of $\lfloor |2\text{-core}|/2 \rfloor$, no d -quotient correction to discover. The order is too shallow to carry information.
- **Littlewood governs the value, not the order.** The classical d -core / d -quotient factorisation does control the *fiber* $G_\lambda(\zeta_d)$ — the value of the polynomial at the root of unity — and this is where the factorisation lift genuinely lives (cf. the 2-core entry of the program, where $G_\lambda(\zeta_d)$ matches an Albion-type plethysm). But “value at ζ_d ” and “order of vanishing at ζ_d ” are different questions, and only the value lifts.
- **The graded order law stands alone at $d = 2$.** It is not the first member of a family; it is a 2-adic phenomenon with no higher- d siblings.

Finally, the computation suggests a clean structural statement to aim for, which the computation verifies but does not prove.

Conjecture 1. *For every $d \geq 3$ and every partition λ , if the d -weight of λ is positive (equivalently, the d -quotient is nonempty), then*

$$\text{ord}_{\zeta_d} G_\lambda = 0.$$

That is, $\Phi_d(q) \mid G_\lambda(q)$ forces λ to be a d -core, and even then the order is exactly 1. (Verified for all $|\lambda| \leq 12$ and $d \in \{3, \dots, 8\}$.)

A proof would presumably proceed through the same fake-degree / cyclotomic-sieve machinery that closed the $d = 2$ law, but run at a d -th root of unity, where the absence of mod- d structure in $s(T)$ should manifest as the impossibility of a double zero. That is the natural next target if one wishes to upgrade the present observation to a theorem.